

PRODUCT REQUIREMENT DOCUMENT

Agentic Claims Review & Denial Prevention Copilot

AI powered claim review, policy intelligence, and denial risk mitigation for revenue cycle teams

PRD v1.1 | Product Strategy & MVP Definition | Jigger Shah | June 11, 2026

Portfolio concept. Built on public data sources and synthetic claims only. No PHI. Not affiliated with any payer, provider, or employer.

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1. Executive Summary

Healthcare providers lose billions of dollars every year to claim denials that never needed to happen. Industry analyses put initial denial rates at 10 to 15% of submitted claims, and Change Healthcare's Denials Index found that roughly 86% of denials are potentially avoidable.. Most of the loss is not from unwinnable disputes. It comes from preventable errors: bundled code pairs billed together, procedures submitted without a covered diagnosis, missing modifiers, and documentation gaps that surface only after the payer rejects the claim.

This PRD defines an AI copilot that moves denial management upstream. Before a claim is submitted, a set of specialized agents validates coding, checks coverage policy, reviews documentation completeness, and produces a denial risk assessment with specific fixes. Every recommendation carries the evidence behind it: the NCCI edit, the NCD/LCD section, the payer policy excerpt. A billing specialist makes the final call on every claim. The system recommends; the human decides.

The MVP is intentionally narrow: a pre-submission review workspace built on free public data sources (NPI Registry, CMS coverage data, NCCI edit files, ICD-10 and HCPCS references) and synthetic claims. It is designed to prove three things: that agentic review catches preventable denial risks, that retrieval grounded explanations earn specialist trust, and that a human-in-the-loop workflow fits how revenue cycle teams actually work. The market is moving the same direction: enterprise agentic platforms launched denial prevention and prior authorization solutions in early 2026, validating pre-submission intelligence as the next layer of revenue cycle automation. [Link 1](#) and [Link 2](#)

2. Problem & Market Context

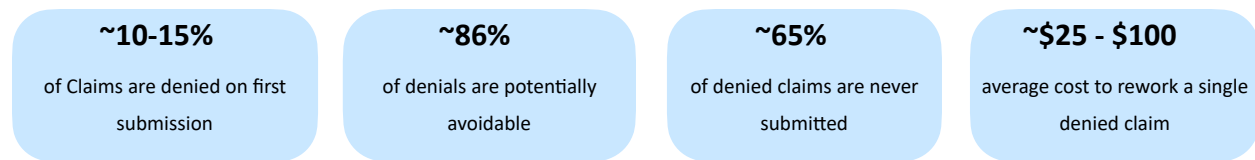


Figure 1. The economics of preventable denials. Sources: Premier (2025), Change Healthcare 2020 Denials Index, MGMA. [Link 1](#) and [Link 2](#).

Denials are a volume problem and a workflow problem at the same time. Premier's 2025 survey found denial rates holding near 15% for private payers, with the cost of fighting denials jumping 23% year over year to \$25.7B, about \$57 per claim before labor. Reworking a single denied claim costs ~\$25 for a physician practice and over ~\$100 for a hospital claim. And because rework is expensive and slow, ~65% of denied claims are never resubmitted at all. That is revenue earned and then abandoned.

Why current workflows fail

- **Knowledge is fragmented.** The answer to “will this claim deny” lives across NCCI edit tables, payer medical policies, LCDs and NCDs, modifier rules, and tribal knowledge held by senior coders.
- **Review is reactive.** Most denial management starts after the 835 comes back with a CARC code. By then the claim has already aged 30 to 45 days.
- **Claim scrubbers are black boxes.** Clearinghouse and EHR edits flag errors but rarely explain the policy basis or suggest the fix, so specialists either trust blindly or re-research manually.
- **Medical necessity is the gap rules can't reach.** Bundling and validity errors are lookups; medical necessity denials (CARC 50) require interpreting coverage policy against the specific diagnosis and procedure pair. That reasoning lives in LCD and NCD text, not edit tables, which is why scrubbers miss it and senior coders spend the most time on it.
- **Expertise does not scale.** Denial pattern knowledge concentrates in a few senior staff. Turnover resets the institution's memory.

Why now

Three things changed.

1. Retrieval augmented generation makes it practical to ground AI output in specific policy text rather than model memory.
2. Agentic orchestration patterns allow specialized validators to run in parallel and hand findings to a reasoning layer, mirroring how a strong coding team actually divides the work.
3. The underlying reference data (NCCI edits, coverage determinations, code sets) is public and machine readable. The raw material for explainable denial prevention has been free all along; what was missing was a product that assembles it into a workflow specialists trust.

3. Vision

Expert-level review on every claim, before it leaves the building — AI does the research, humans make the call.

The copilot gives revenue cycle teams the combined judgment of a senior coder, a payer policy analyst, and a denial specialist, applied to 100 percent of claims instead of the small sample a human team can touch, with every recommendation backed by cited evidence and every final decision made by a human.

Positioning in one sentence

An explainable, agentic pre-submission copilot that prevents denials before they happen, rather than a scrubber that flags errors without context or an analytics tool that explains denials after the revenue is already at risk.

Strategic Value

The long-term opportunity extends beyond denial prevention. Claims review is the entry point to a Revenue Cycle Agent Platform: the same governed agent architecture, policy intelligence layer, and evidence retrieval framework that reviews a claim can drive prior authorization, appeals, coding assistance, and documentation intelligence. Build the decision layer once, with governance designed in, and each new workflow is an extension rather than a new product. The phased roadmap in Section 16 follows exactly this sequence.

4. Target Users

User Persona	Role in workflow	What they need from the product
Revenue Cycle Specialist (primary)	Reviews flagged claims, applies fixes, owns submission	Fast triage, clear fixes, evidence they can verify in under a minute
Medical Coder (primary)	Validates code selection, modifiers, and bundling	NCCI and modifier logic with the specific edit citation, not a generic flag
Revenue Cycle Manager	Owens first-pass rate and team throughput	Queue visibility, override patterns, denial risk trends
Compliance Lead	Owens audit readiness and coding integrity	Full decision trail: what the AI recommended, what the human did, and why
Practice Administrator / Finance	Owens cash flow and cost to collect	Measurable lift in first-pass acceptance and days in A/R

Primary persona snapshot

Maria, Revenue Cycle Specialist at a 40-provider multi-specialty group. She works a queue of 80 to 120 claims a day. Today she relies on the clearinghouse scrubber plus her own memory of which payers deny what. When the scrubber flags something, she spends 5 to 15 minutes hunting through payer portals and coding references to confirm whether the flag is real. Her core question for any AI tool: “Show me why, and let me decide.”

5. Product Principles

Principle	What it means in practice
1. Assist, don't replace	The copilot never submits, corrects, or denies anything autonomously. Every action is a recommendation pending human decision.
2. Evidence before recommendation	No finding ships without a citation: the NCCI edit number, the LCD section, the policy excerpt. If the system cannot cite it, it does not say it.
3. Deterministic before generative	Code validity, NPI status, NCCI pairs, and MUE limits are rule lookups, not LLM calls. Generative reasoning is reserved for synthesis, coverage interpretation, and documentation review where rules cannot reach.
4. Human accountability	A named specialist accepts, modifies, or overrides every recommendation. Overrides require a reason and feed back into evaluation.
5. Governance by design	Audit trail, confidence thresholds, and escalation paths are MVP scope, not a later phase. Trust is a launch requirement.

6. MVP Scope & Prioritization

The MVP is a single-claim review workspace. A user enters or uploads a claim, the agents review it, and the user works through findings to a submit-ready decision. Prioritization below reflects what is required to prove the core value (P0), what strengthens trust and demo quality (P1), and what is deferred (P2).

Priority	Capability	Notes
P0	Claim intake form: payer, NPI, CPT/HCPCS, ICD-10, modifiers, place of service, units	Manual entry plus CSV upload of synthetic claims
P0	Deterministic validation: NPI lookup, code validity, NCCI PTP pairs, MUE limits	Rule layer runs before any LLM call
P0	Coverage check against CMS NCD/LCD data for Medicare scenarios	Diagnosis-to-procedure medical necessity fit
P0	Denial risk assessment with severity, recommended fix, and citation per finding	The core agentic output
P0	Accept / modify / override workflow with required override reason	Human decision is the unit of record
P0	Audit log: inputs, agent findings, citations, human decision, timestamps	Exportable per claim
P1	Confidence score per finding with low-confidence escalation flag	Drives the trust narrative
P1	Documentation review agent on pasted clinical note text (synthetic)	Specificity and required-element gaps
P1	Batch review of an uploaded claim file with a triage queue view	Shows workflow scale
P2	Commercial payer policy corpus beyond CMS	Public payer medical policies, indexed for retrieval
P2	Denial pattern analytics across reviewed claims	Trend view for managers

Out of scope for MVP

EDI 837/835 integration, real claim submission, eligibility verification (270/271 is commodity clearinghouse functionality and belongs at registration, not claim review), appeals workflows, prior authorization workflows, automated correction without human action, production payer connectivity, and any use of real patient data. These are deliberate exclusions: the MVP proves decision quality and trust, not pipes.

7. User Stories & Acceptance Criteria

Story 1: Pre-submission risk review (P0)

As a revenue cycle specialist, I want every claim reviewed for denial risk before submission, so that I fix problems while the fix is cheap.

- Given a complete claim, the system returns findings in under 30 seconds.
- Each finding includes severity (high / medium / low), the specific issue, a recommended fix, and at least one citation to source policy or edit data.
- A claim with no findings is explicitly marked “no denial risks identified” with the checks that were run listed.

Story 2: Evidence verification (P0)

As a medical coder, I want to see the exact policy basis for every flag, so that I can verify it instead of trusting a black box.

- Clicking a citation shows the source excerpt (edit table row, LCD section, policy text) and its effective date.
- Findings without a verifiable source are suppressed, never shown as unsourced assertions.

Story 3: Override with accountability (P0)

As a specialist, I want to override a recommendation when my judgment differs, so that the AI never blocks my work.

- Override requires a free-text reason and is captured in the audit log with user, timestamp, and the recommendation overridden.
- Accepted, modified, and overridden recommendations are all query-able for evaluation.

Story 4: Audit trail (P0)

As a compliance lead, I want a complete decision trail per claim, so that AI-assisted review strengthens audit readiness instead of threatening it.

- Per-claim export includes inputs, every agent finding with citations, confidence, and the human decision sequence.

8. Worked Example: One Claim, End to End

Synthetic claim, illustrative codes. A specialist at a primary care group enters the following claim for a Medicare patient seen yesterday.

Field	Value
Payer	Medicare Part B
Rendering NPI	1234567890 (synthetic)
CPT / HCPCS	99214 (established patient E/M), 80053 (comprehensive metabolic panel), 80048 (basic metabolic panel), 36415 (venipuncture)
ICD-10	Z00.00 (general adult exam without abnormal findings)
Modifiers	None
Place of service	11 (office)

What the copilot returns (under 30 seconds)

Finding	Agent	Severity	Recommendation and evidence
Bundled code pair: 80048 is a component of 80053	Coding Validation	High	Remove 80048. NCCI PTP edit lists 80048 as bundled into 80053 with modifier indicator 0: no modifier bypass allowed. Citation: NCCI PTP edit table, current quarter.
Diagnosis does not support E/M level billed	Coverage Validation	High	Z00.00 (routine exam, no abnormal findings) conflicts with a problem-oriented 99214. Either bill the appropriate preventive code, or add the problem diagnosis and confirm the note supports moderate MDM. Citation: ICD-10-CM Z00.00 usage notes; E/M guidelines.
Possible missing modifier 25 scenario	Coding Validation	Medium	If a preventive service and a separate E/M both occurred, 99214 needs modifier 25. Confirm intent. Citation: NCCI Policy Manual, modifier 25 guidance.
Documentation not provided	Documentation Review	Low	No note attached. Risk assessment is based on codes only; attach the note to validate medical decision making level.

Denial risk score: HIGH (two high-severity findings). The specialist removes 80048, confirms with the provider that hypertension was addressed, adds I10 as a secondary diagnosis, accepts the recommendations, and marks the claim submit-ready. Total time: about 90 seconds. Without the review, this claim carried two near-certain denial triggers (a CO-97 bundling denial and a medical necessity denial), each worth 30+ days of delay and a rework cycle. The audit log records all of it.

9. Agent Architecture

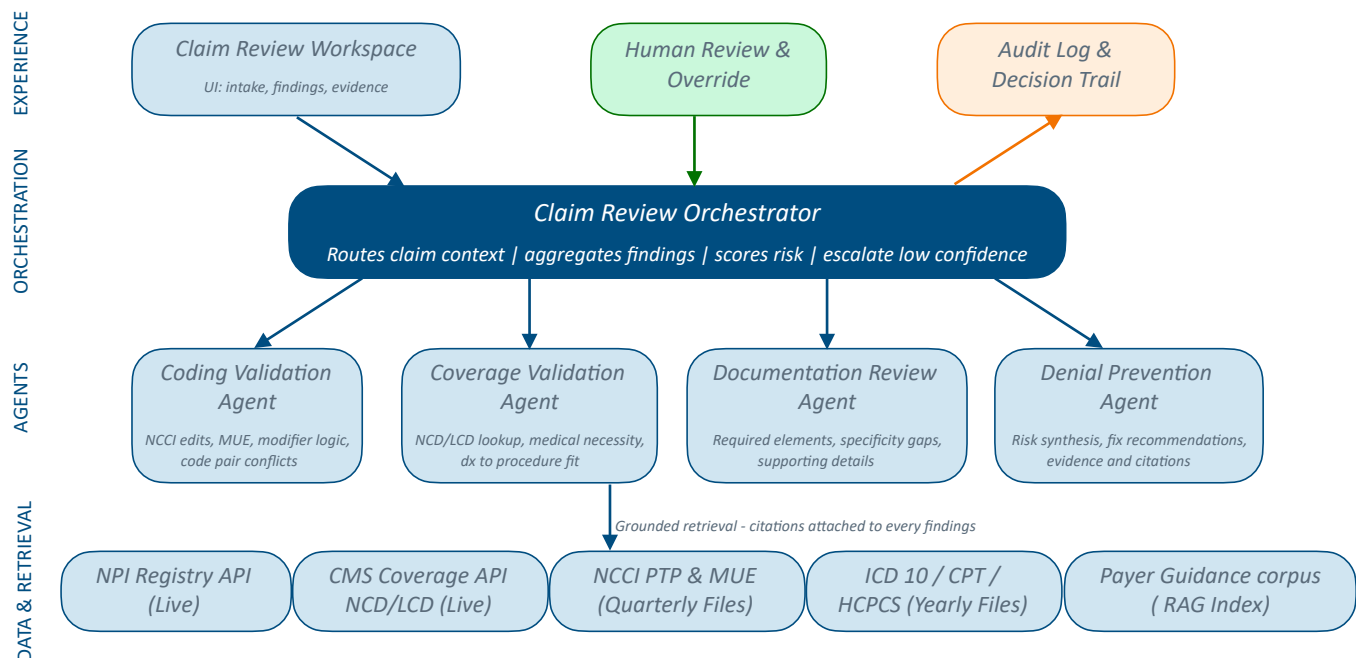


Figure 2. Four specialized agents behind a single orchestrator, grounded in public reference data.

The orchestrator decomposes each claim into validation tasks, runs the deterministic rule layer first, then dispatches agents in parallel. Findings come back as structured objects (issue, severity, fix, citation, confidence) that the Denial Prevention Agent synthesizes into a single risk assessment. Low aggregate confidence routes the claim to a human-first queue rather than presenting uncertain recommendations as confident ones.

Agent	Type of reasoning	Primary sources
Coding Validation	Deterministic lookups plus rule interpretation	NCCI PTP and MUE files, HCPCS/CPT reference data, modifier rules
Coverage Validation	Retrieval grounded reasoning	CMS Coverage API (NCDs, LCDs), diagnosis-to-procedure policy mapping
Documentation Review	LLM analysis of note text	E/M guidelines, specificity requirements, payer documentation rules
Denial Prevention	Synthesis and risk scoring	All agent findings plus denial pattern heuristics

Claims Review Workflow

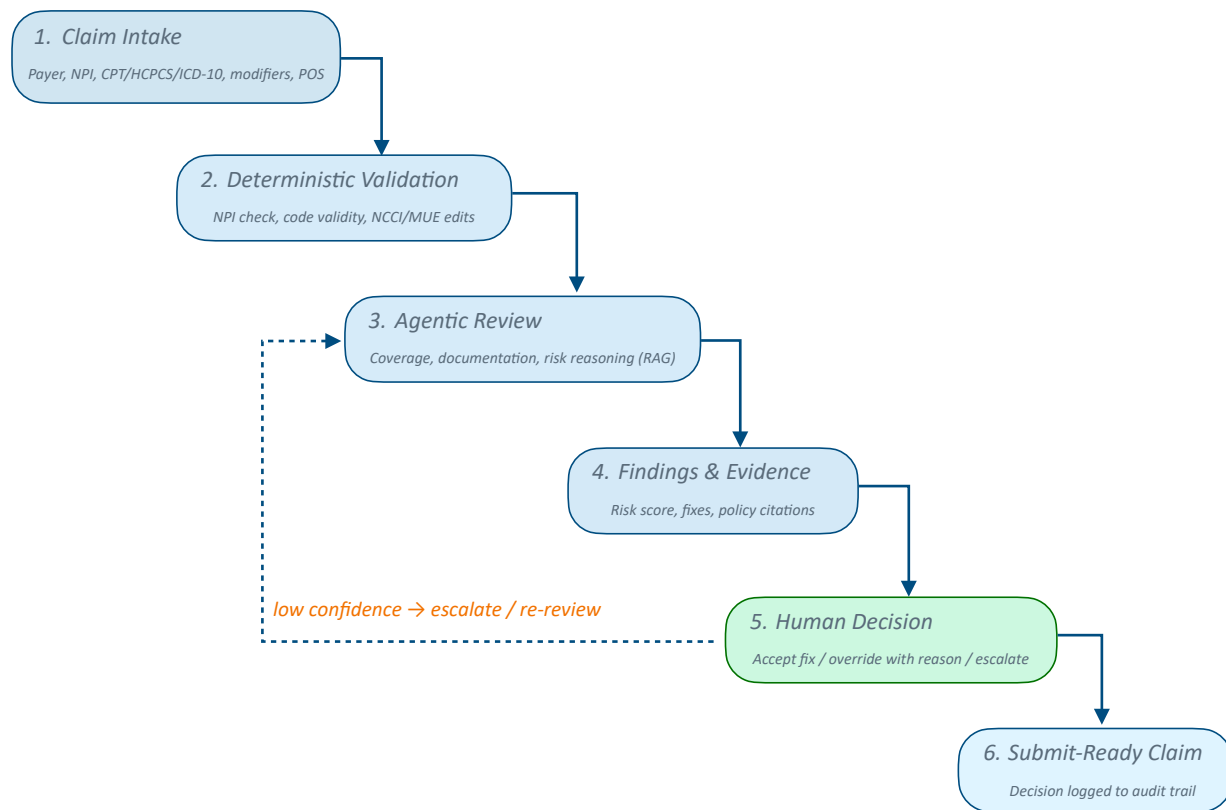


Figure 3. Deterministic checks run before generative reasoning. The human decision is the last step, always.

10. Data Sources

Every source below is free and public. The architectural distinction matters: live APIs are queried at review time; reference files are ingested on their publication cadence and versioned, so every citation can name the edition it came from.

Source	Access type	Refresh cadence	Used for
NPPES NPI Registry API	Live REST API	Real time	Provider identity, taxonomy, active status
CMS Coverage API (MCD)	Live REST API	Real time	NCDs, LCDs, coverage articles for medical necessity
NCCI PTP edits	File download, ingested	Quarterly	Bundled code pair detection, modifier indicators
NCCI MUE tables	File download, ingested	Quarterly	Units-of-service limits per code
ICD-10-CM code set	File download, ingested	Annual (Oct), with updates	Diagnosis validity, specificity, usage notes
HCPCS Level II	File download, ingested	Quarterly	Supply, drug, and service code validity
Public payer medical policies	Scraped/collected corpus, RAG indexed	As published (P2)	Commercial coverage reasoning beyond CMS

11. MVP Technical Approach

Built to be reproducible by anyone with free resources. The stack is deliberately boring everywhere except the agent layer.

- **Frontend and app shell:** Streamlit. Single-claim workspace, findings panel with expandable evidence, audit log view.
- **Rule layer:** Python. NCCI/MUE lookups against ingested quarterly files, code validity checks, NPI API client. No LLM involvement.
- **Agent layer:** LLM API (Claude or equivalent) with structured tool use. Each agent is a constrained prompt plus tools; the orchestrator is a deterministic Python controller, not a free-running loop.
- **Retrieval:** Local vector store (Chroma or FAISS) over LCD/NCD text and policy documents. Citations carry document ID, section, and effective date.
- **Data:** 100 percent synthetic claims generated to exercise known denial patterns. No PHI anywhere in the system.
- **Persistence:** SQLite for claims, findings, decisions, and audit events. Exportable to CSV.

A companion GitHub repository will hold the codebase structure, a README, synthetic data generators, and setup instructions, so the project doubles as a public reference implementation of governed agentic AI in revenue cycle.

12. Responsible AI & Governance

Control	Implementation in MVP
Human in the loop	No system action changes a claim. Specialists accept, modify, or override every recommendation; overrides require a reason.
Grounded outputs	Coverage and policy findings must cite retrieved source text. Unverifiable findings are suppressed by design.
Confidence-based escalation	Aggregate confidence below threshold routes the claim to human-first review with the uncertainty stated plainly.
Auditability	Immutable per-claim log of inputs, findings, citations, confidence, and human decisions, with timestamps.
PHI protection	Synthetic data only in MVP. The architecture assumes a future deployment boundary where claim data never leaves the customer environment.
Evaluation and monitoring	Golden set of synthetic claims with known correct findings; precision and recall tracked per agent per release. Override rates monitored as a drift signal.

13. Alternatives & Competitive Landscape

Alternative	What it does well	Where the copilot differs
Clearinghouse / EHR claim scrubbers (Waystar, Availity, EHR-native edits)	High-volume rule edits at submission time	Edits flag; they rarely explain or recommend. The copilot pairs every flag with policy evidence and a specific fix, earlier in the workflow.
Denial analytics platforms (MDaudit, payer analytics suites)	Retrospective denial trends and root cause reporting	Analytics explain last quarter's denials. The copilot prevents this week's, claim by claim, pre-submission.
RCM outsourcing / coding services	Human expertise at scale, for a percentage of collections	The copilot scales the same expertise across 100 percent of claims at software cost, while keeping decisions in-house.
Emerging agentic RCM automation (incl. UiPath-class agentic platforms)	End to end task automation across RCM systems	Complementary, not competitive. UiPath's ViVE 2026 launch (denial prevention and resolution, prior authorization with Genzeon) validates the problem space at the enterprise orchestration layer. This PRD is a transparent reference implementation of the decision layer underneath: evidence-grounded findings, public data only, every recommendation citable. The two meet where automation needs explainable judgment to act safely.

Positioning summary: scrubbers are rules without reasons, analytics are reasons without prevention. The copilot is prevention with reasons.

14. Success Metrics

Metric	Definition	MVP target / measurement
Finding precision	Share of agent findings judged correct against the golden synthetic set	≥ 90% on golden set before demo
Finding recall	Share of seeded denial risks the system catches	≥ 85% on golden set
Medical necessity determination accuracy	Coverage findings matching the governing LCD/NCD on the golden set	≥ 90%; tracked separately because it's the highest-difficulty reasoning class
Citation coverage	Share of findings carrying a verifiable source citation	100% (suppress otherwise)
Recommendation acceptance rate	Accepted plus modified, over total recommendations	Tracked from day one; the core trust signal
Average review time per claim	Intake to submit-ready decision	≤ 2 minutes on standard claims
Escalation rate	Claims routed to human-first queue on low confidence	Tracked; expect 10–20% early, declining
First-pass acceptance (north star)	Clean claim rate after adoption	Production metric, modeled in MVP via prevented-denial count

15. Risks & Open Questions

Risk / question	Mitigation / decision needed
LLM asserts policy that retrieval did not return	Hard rule: no citation, no finding. Coverage reasoning constrained to retrieved text. Golden set regression on every change.
Reference data staleness (quarterly NCCI, annual ICD-10)	Version every ingested file; show edition in citations; block review if data is past expiry.
Alert fatigue: too many low-value findings	Severity tiers, suppress informational noise by default, track dismissal patterns per finding type.
Commercial payer policies are inconsistent and unstructured	MVP stays CMS-first where data is clean. Commercial corpus is P2 with explicit per-payer confidence.
Open: where does this sit in the stack long term?	Standalone workspace vs. embedded decision service inside an EHR, clearinghouse, or agentic automation platform. The API-first agent layer keeps both paths open.
Open: what override rate signals a trust problem vs. healthy skepticism?	Hypothesis: sustained overrides above 30% on high-severity findings means precision work is needed. Validate with usage.
Open: eligibility denials are the largest denial category, but verification requires payer connectivity the MVP deliberately excludes	In production, the copilot consumes eligibility status as an input signal, flagging claims with stale or missing verification, rather than performing the check itself. Verification stays at registration where it belongs.

16. Roadmap

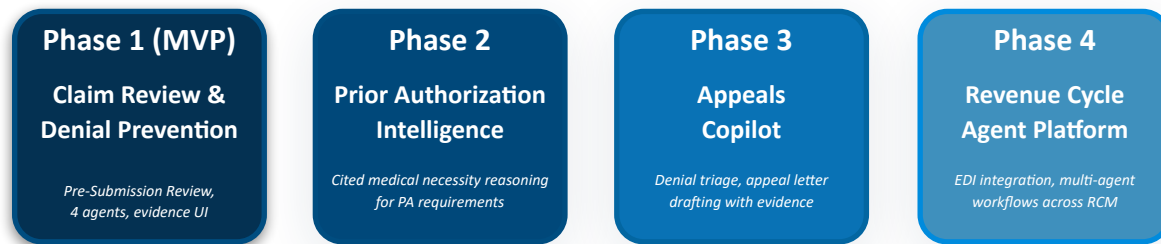


Figure 4. Each phase reuses the same orchestration, retrieval, and governance foundation.

- **Phase 1** proves explainable pre-submission review.
- **Phase 2** extends the coverage agents to prior authorization requirement detection. Enterprise platforms including UiPath shipped agentic prior auth in early 2026, which confirms the demand; this phase focuses on the piece those platforms route around, transparent medical necessity reasoning with cited policy evidence.
- **Phase 3** turns the same evidence engine around to draft appeals for denials that slip through.
- **Phase 4** connects to live claim pipelines (EDI 837/835) and opens the agent layer as a platform other RCM automation can call.

17. Product Narrative

Denial management has been reactive for thirty years: submit, wait, get denied, investigate, rework, resubmit, and abandon what is too expensive to fight. The technology to invert that loop now exists, and the reference data to ground it has been public all along. This product applies agentic AI where it is strongest, parallel specialized review with cited evidence, and keeps humans where they are essential, in the decision. The result is not an autonomous biller. It is a copilot that makes every specialist as sharp as the best one on the team, on every claim, before the claim leaves the building.

Appendix note: All claims, NPIs, and clinical scenarios in this document are synthetic and illustrative. Industry statistics are drawn from published analyses by Premier (2025), the Advisory Board, MGMA, and Change Healthcare, and are cited as directional context. This is an independent portfolio concept by Jigger Shah.